

Efficient Market Model

Security Diversification

- Equally likely Events

	Equity		Bond	
Scenario	Return	Variance	Return	Variance
Bearish	-0.1622	0.0005	-1.0854	0.0003
Normal	0.8214	0.0008	-0.1089	0.0008
Bullish	0.2608	0.0009	0.1172	0.0003
Expected Return	0.3067		-0.3590	
Variance	0.0008		0.0005	
Standard Deviation	0.0274		0.0215	

- Expected return = $\frac{1}{3} \times (\text{Bearish} + \text{Normal} + \text{Bullish})$
- Variance = $\frac{1}{3} \times (\text{Bearish} + \text{Normal} + \text{Bullish})$
- Standard Deviation = Square root of variance

Portfolio Return

- Equally weighted portfolio (weights: 50:50)

Scenario	Equity	Bond	Portfolio Return
Bearish	-0.1622	-1.0854	-0.6238
Normal	0.8214	-0.1089	0.3562
Bullish	0.2608	0.1172	0.1890
Expected Return	0.3067	-0.3590	-0.0262

- $$E(r_{\text{Portfolio}}) = (w_{\text{Equity}} \times r_{\text{Equity}}) + (w_{\text{Bond}} \times r_{\text{Bond}})$$

Portfolio Risk

- Portfolio Risk Vs. Return

$$\text{Portfolio return} = E(r) = \sum_{i=1}^n \omega_i r_i$$

$$\text{Portfolio risk} = \sqrt{\sum_{i=1}^n \omega_i^2 \sigma_{ii} + \sum_{i=1}^n \sum_{j=1}^n \omega_i \omega_j \sigma_{ij}}$$

where ω_i – Weight of a security; r_i – Security return;

n – Number of securities; σ_{ij} – Standard deviation

$$\sum_{i=1}^n \omega_i = 1$$

Computation of Portfolio Risk

- Variance Covariance Matrix

<i>Security</i>	<i>1</i>	<i>2</i>	<i>3</i>		<i>n</i>
1	$\omega_1 \omega_1 \sigma_{11}$	$\omega_1 \omega_2 \sigma_{12}$	$\omega_1 \omega_3 \sigma_{13}$		$\omega_1 \omega_n \sigma_{1n}$
2	$\omega_2 \omega_1 \sigma_{21}$	$\omega_2 \omega_2 \sigma_{22}$	$\omega_2 \omega_3 \sigma_{23}$		$\omega_2 \omega_n \sigma_{2n}$
3	$\omega_3 \omega_1 \sigma_{31}$	$\omega_3 \omega_2 \sigma_{32}$	$\omega_3 \omega_3 \sigma_{33}$		$\omega_3 \omega_n \sigma_{3n}$
.....					
<i>n</i>	$\omega_n \omega_1 \sigma_{n1}$	$\omega_n \omega_2 \sigma_{n2}$	$\omega_n \omega_3 \sigma_{n3}$		$\omega_n \omega_n \sigma_{nn}$

$$\sigma_p^2 = \sum_{i=1}^n \omega_i^2 \sigma_{ii} + \sum_{i=1}^n \sum_{j=1}^n \omega_i \omega_j \sigma_{ij}$$

$$\sigma_p = \sqrt{\sum_{i=1}^n \omega_i^2 \sigma_{ii} + \sum_{i=1}^n \sum_{j=1}^n \omega_i \omega_j \sigma_{ij}}$$

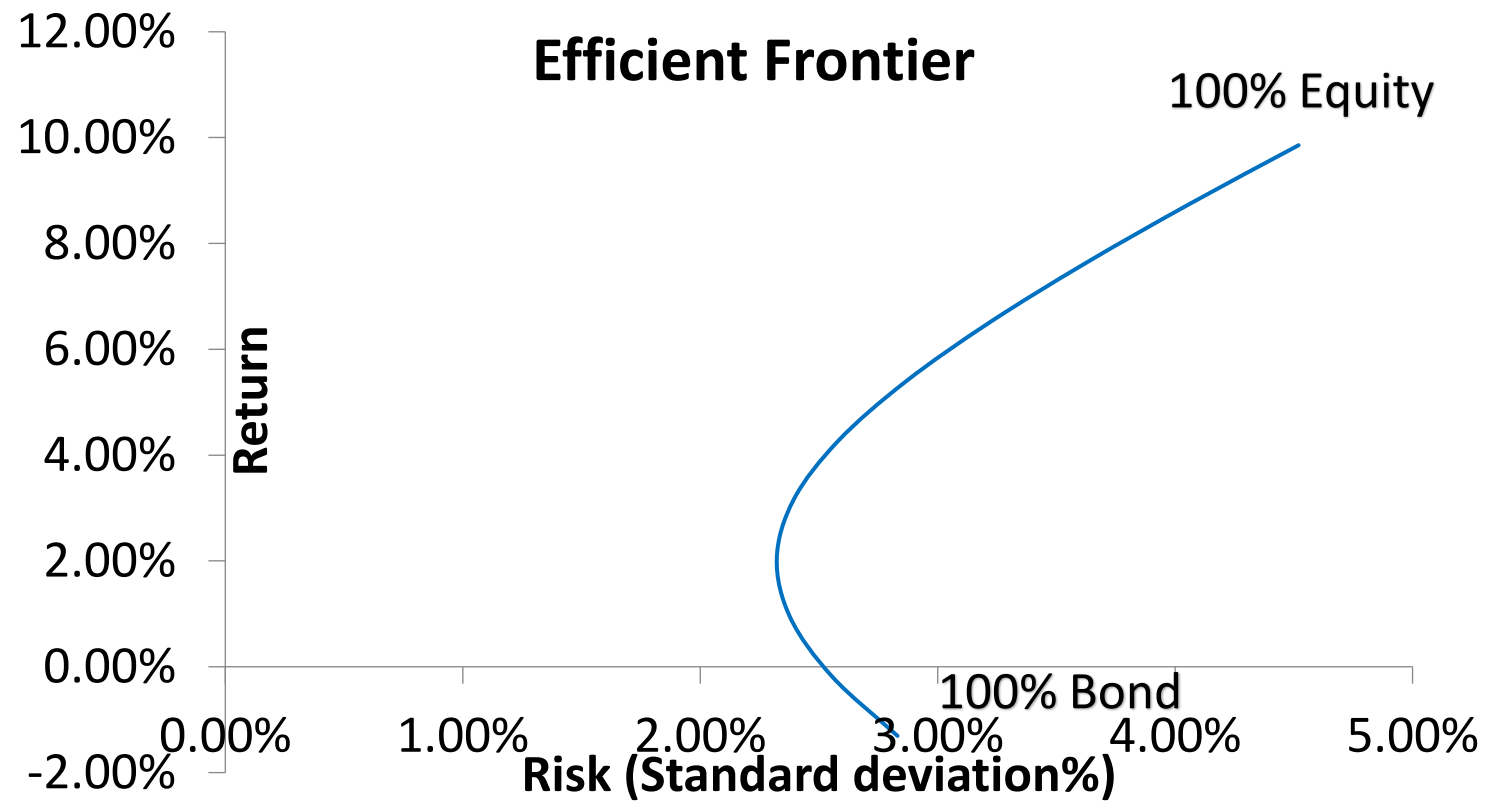
Portfolio Risk

- Two security portfolio
- $\sigma_p^2 = (\omega_i \sigma_i)^2 + (\omega_j \sigma_j)^2 + 2(\omega_i \sigma_i)(\omega_j \sigma_j)\rho_{ij}$
- ρ_{ij} is the correlation coefficient between security i and j

Scenario	Equity Variance	Bond Variance	Correlation	Portfolio Variance	Portfolio Standard Deviation
Bearish	0.0005	0.0003	-0.1165	0.000115449	0.010745
Normal	0.0008	0.0008	-0.1165	0.00021404	0.01463
Bullish	0.0009	0.0003	-0.0509	0.000249107	0.015783

Efficient Frontier

- Normal Scenario



Bond Market Vs. Sensex Return

Monthly Return	Bond	Sensex	Bond /Sensex
Average	-0.0011	0.0082	
Variance	0.0008	0.0020	
Standard Deviation	0.0283	0.0452	
Correlation			-0.0692
Covariance			-0.0001

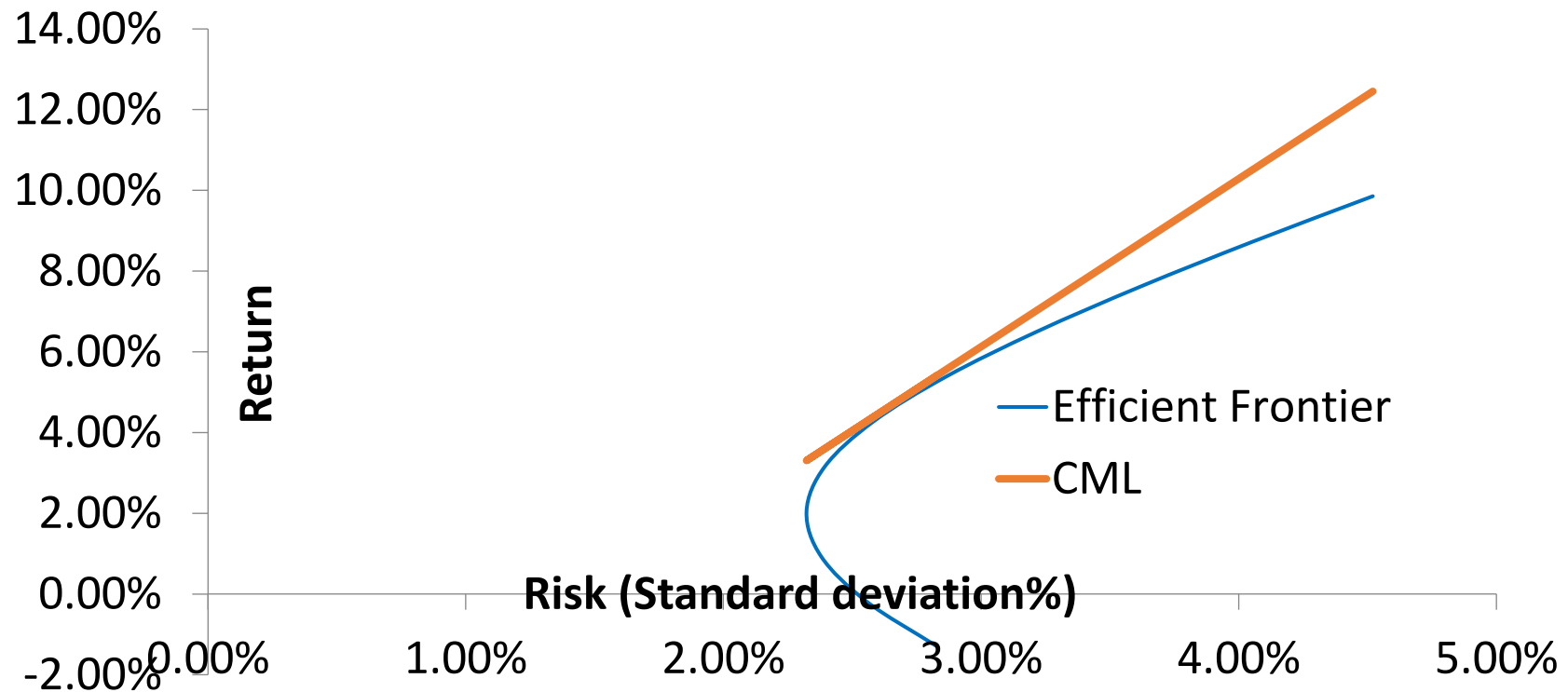
Capital Market Line

Risk free rate:
 0.0053
 Annualized Risk
 free rate: 6.36%
 Slope: 4.162
 Portfolio return
 = f (Portfolio
 risk)

Bond weights	Equity weights	Mean Return	Variance	Standard Deviation	CML
0%	100%	9.86%	0.0020	4.52%	12.45%
10%	90%	8.74%	0.0016	4.06%	10.53%
20%	80%	7.62%	0.0013	3.62%	8.71%
30%	70%	6.51%	0.0010	3.22%	7.04%
40%	60%	5.39%	0.0008	2.87%	5.57%
50%	50%	4.27%	0.0007	2.58%	4.39%
60%	40%	3.16%	0.0006	2.39%	3.60%
70%	30%	2.04%	0.0005	2.32%	3.31%
80%	20%	0.93%	0.0006	2.38%	3.54%
90%	10%	-0.19%	0.0007	2.56%	4.28%
100%	0%	-1.31%	0.0008	2.83%	5.42%

Capital Market Line

- Capital Market Line = (Standard Deviation x Slope) – (Risk Free Rate)



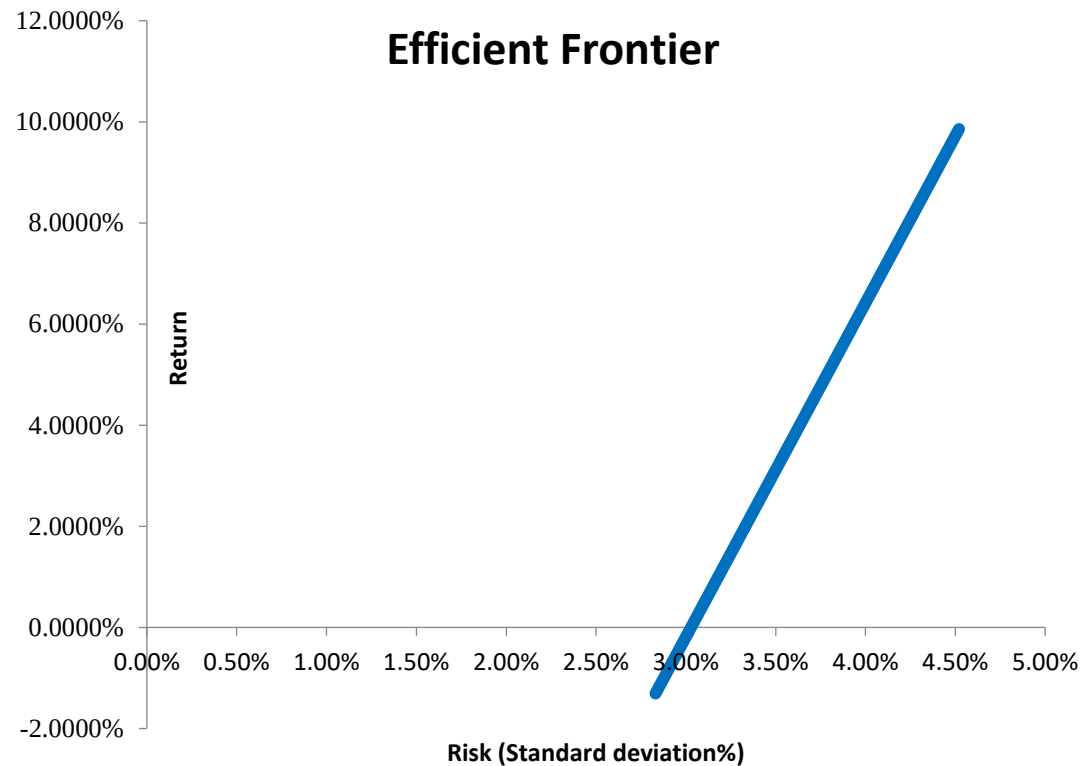
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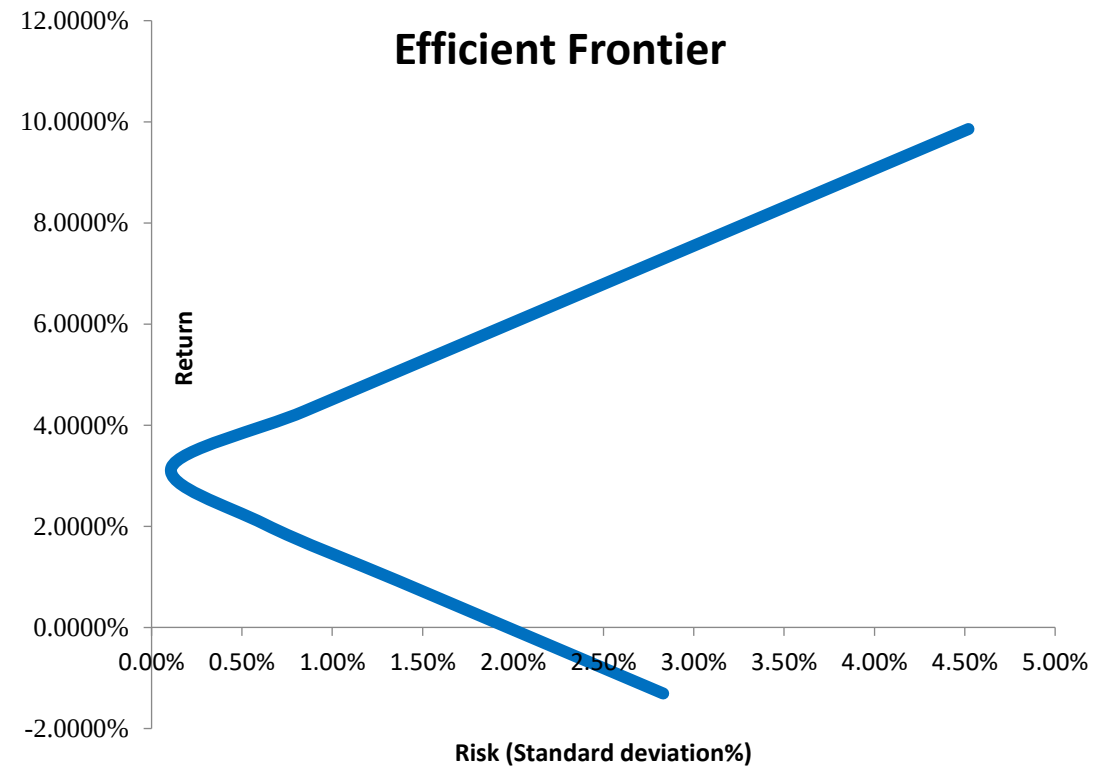
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Varying Correlation Coefficient

Correlation: 1

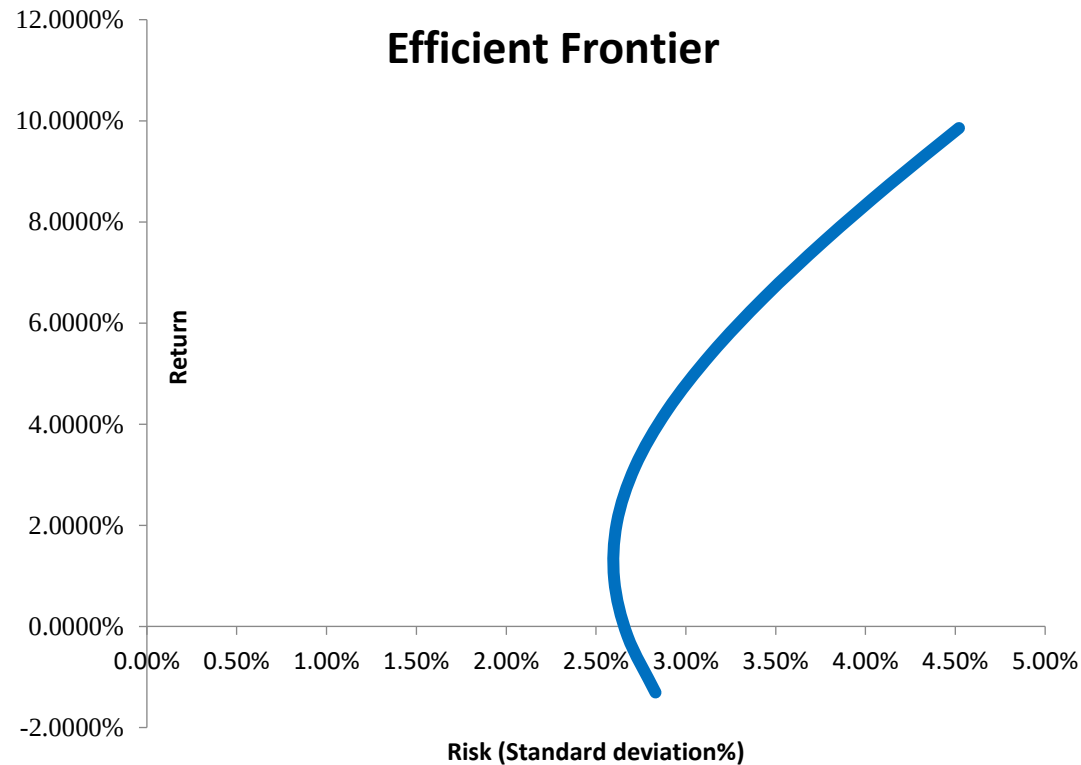


Correlation: -1

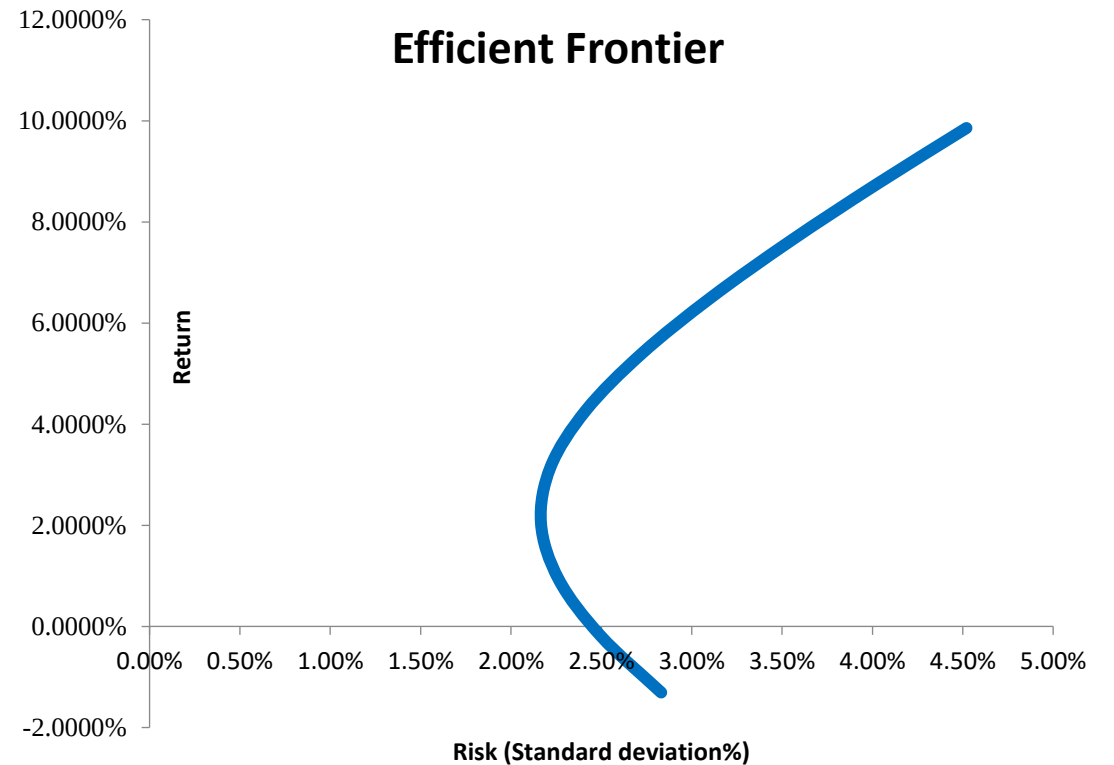


Varying Correlation Coefficient

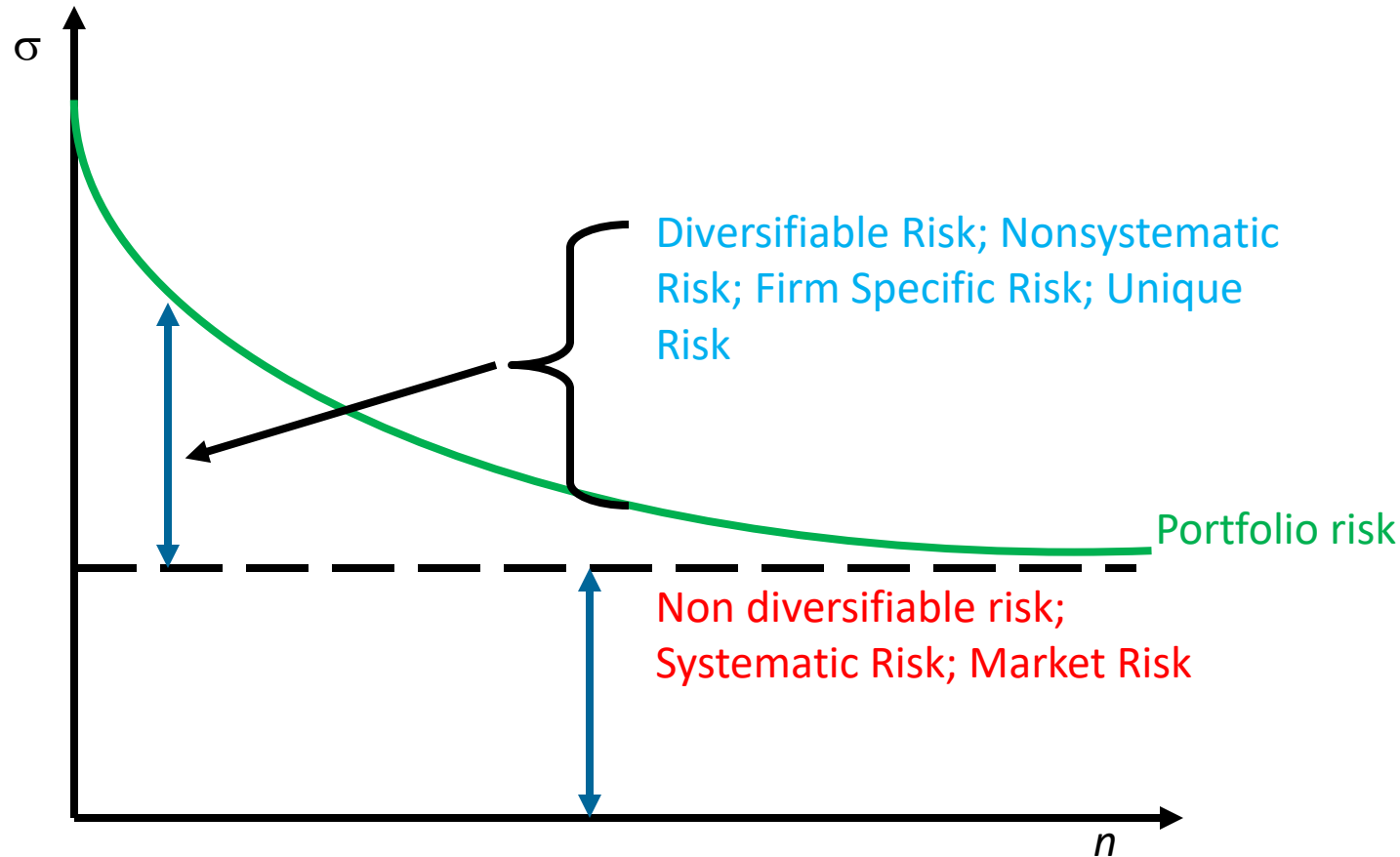
Correlation: 0.2



Correlation: -0.2



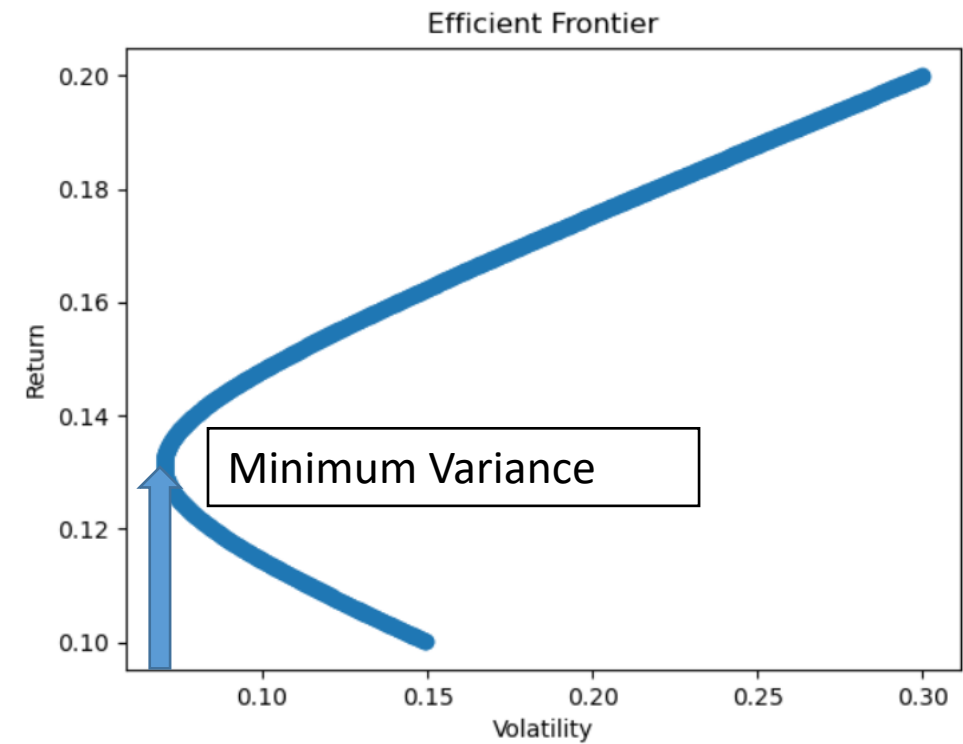
Portfolio Risk and Number of Securities



- In a large security portfolio, variances are effectively diversified, but covariances are not diversified.
- Diversification can eliminate some, but not all of the risk of individual securities.

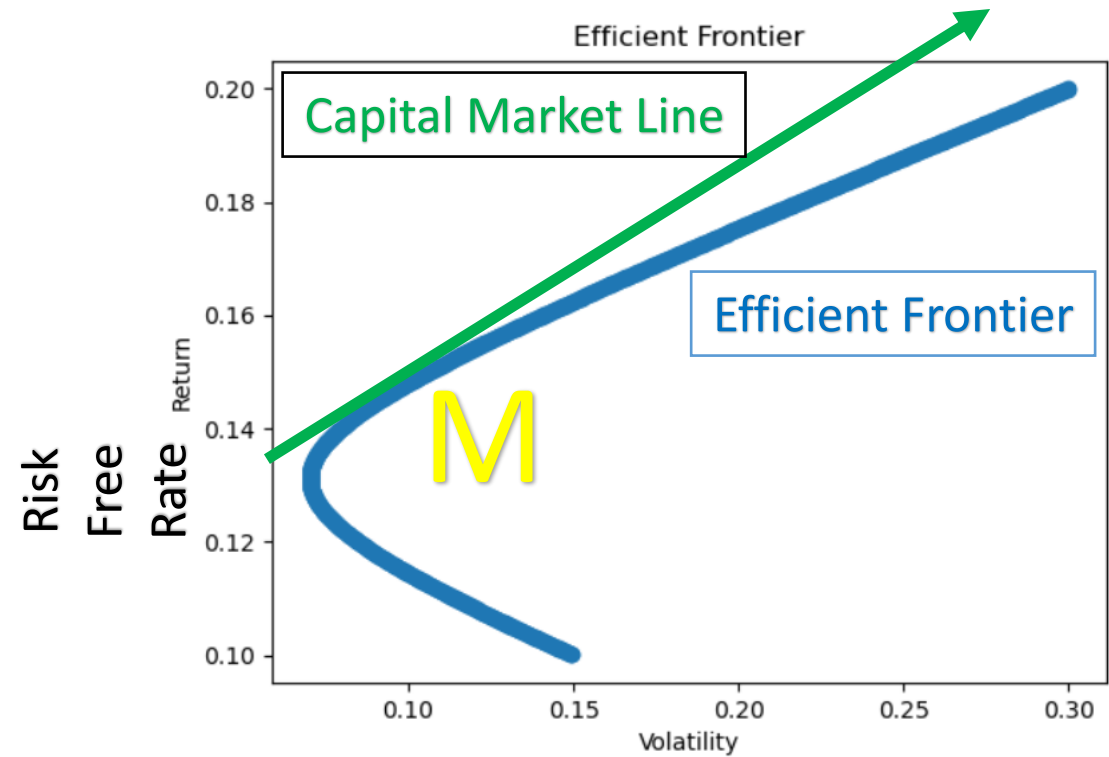
Efficient Portfolio of Large Securities Portfolio

- Efficient Frontier is recognized with the minimum variance portfolio
- Any other portfolio below the minimum variance portfolio provides a lower return for the risk associated with the portfolio.
- Portfolios represented above the minimum variance portfolio is the efficient frontier



Optimal Portfolio with a Risk Free Asset

- Investors can allocate weights for Risk free rate (Treasury bills) and Portfolios in the Efficient Frontier
- Investors have the same CML since they have the same optimal risky portfolio given the risk-free rate
- Market rate and market volatility (M) are the same for CML and EF



Capital Market Line

- Points on the CML except the Market Portfolio dominate all points on the Markowitz efficient set (provide a higher expected return for any given level of risk).
 - Investors should invest in the same risky portfolio (M), and then lend or borrow at the risk-free rate depending on their risk preferences.
- Portfolios on the CML are a combination of two assets
 - Risk-free asset,
 - Market Portfolio

$$\sigma^2(r_p) = x_{r_F}^2 \sigma^2(r_F) + x_M^2 \sigma^2(r_M) + 2 x_{r_F} x_M \rho_{r_F, M} \sigma(r_F) \sigma(r_M)$$

However, since $\sigma(r_F) = 0$ (Risk-Free)

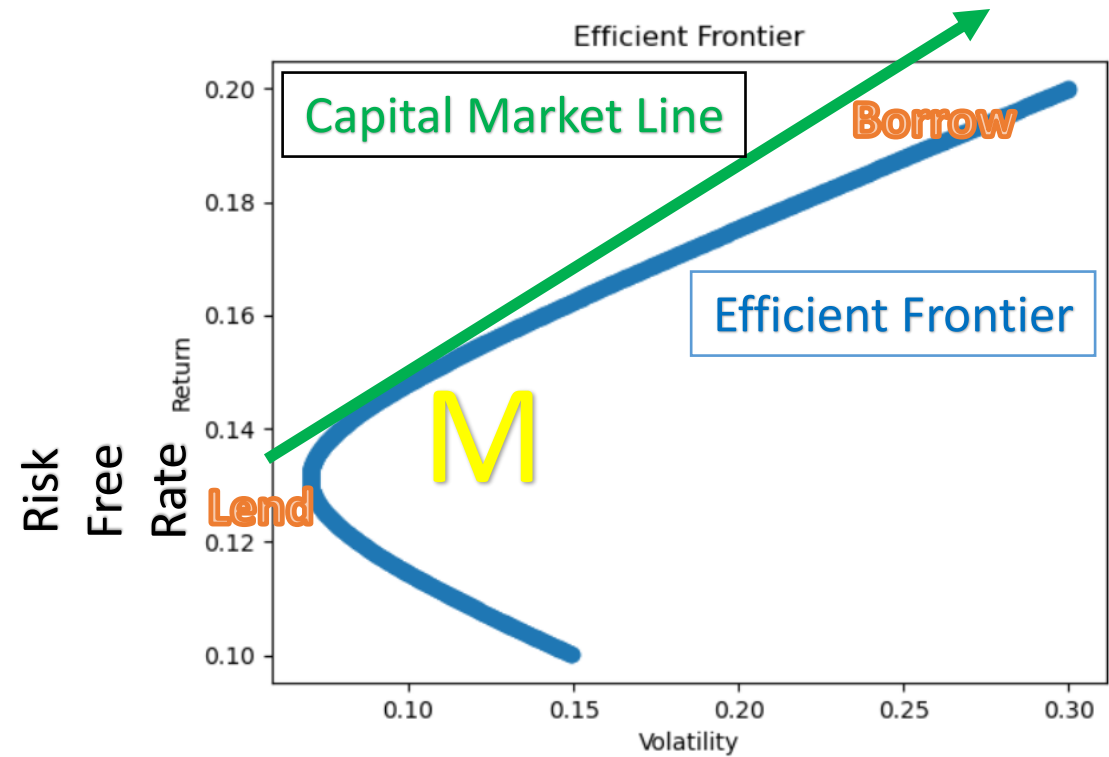
$$\sigma^2(r_p) = x_M^2 \sigma^2(r_M) \text{ and } \sigma(r_p) = x_M \sigma(r_M)$$

Risk of CML

- Since $\sigma(r_p) = x_M \sigma(r_M)$, all portfolios that lie on the CML are perfectly positively correlated with the Market Portfolio
- 100% of the variance in the portfolio's returns is explained by the variance in the market's returns, when the portfolio lies on the CML.
- Efficient frontier assumes correlation between assets is constant, however this may not hold in a volatile market.
- Over-diversification might force investors to hold very small proportions with large securities that eventually may not be able to take advantage of potential growth opportunities.

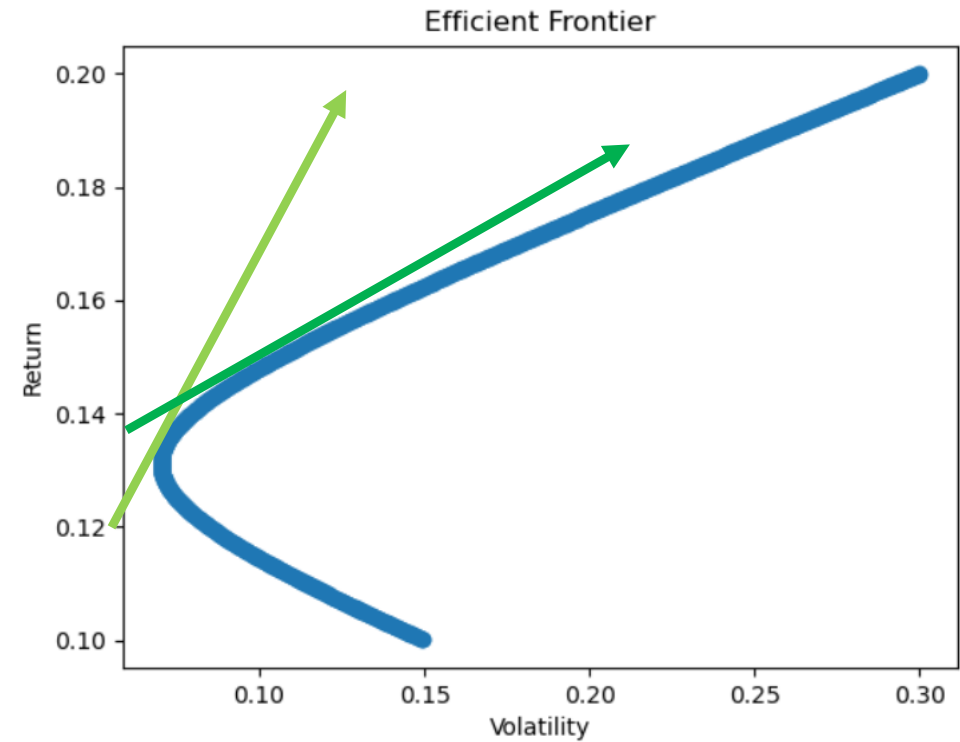
Optimal Portfolio with a Risk Free Asset

- CML above Efficient Frontier implies investors borrow at risk free rate to invest in portfolio since borrowing rate is cheaper
- CML below Efficient Frontier implies investors invest a part of their investment in risk-free asset



Separation Property

- Investor risk aversion is in their choice of where to stay along the CML; not in their choice of the line.
- Separation property implies that portfolio choice can be separated into two tasks:
 - Determine the optimal risky portfolio,
 - Selecting a point on the CML
- Optimal risky portfolio depends on the risk-free rate as well as the risky assets.



Alternative Portfolio Optimization Models

- Minimum Variance Portfolio
 - Select the proportion of assets that minimize the portfolio volatility.
- Maximum Sharpe Ratio Portfolio
 - Select the proportion of assets that maximize the risk adjusted return.